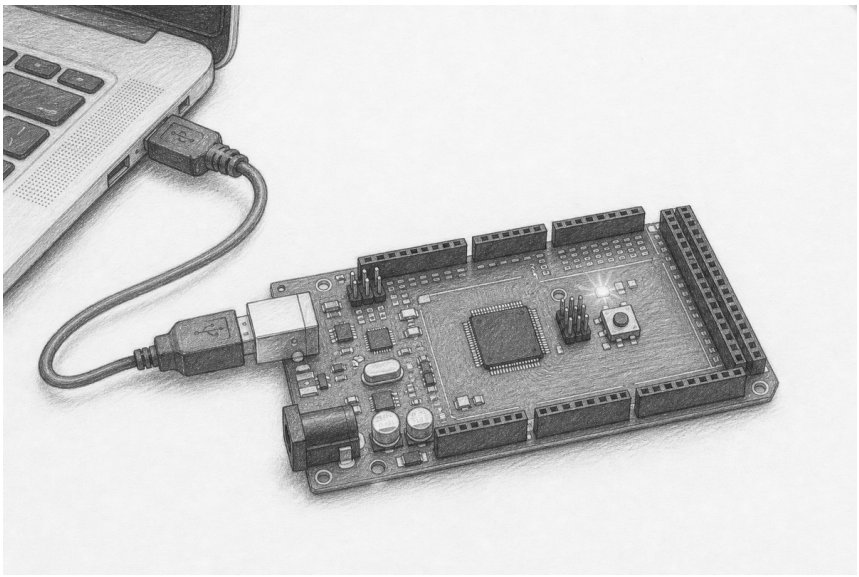
Make state visible

Lesson 001: safe digital output

Predict → claim → command → release



Orientation plate. This graphite view helps you find the Mega 2560, USB connector, and built-in indicator. It is not a pin map. The exact electrical path appears below.

Question	Can an owning C++ object make one hardware state visible, then return its pin to a safe high-impedance state?
Time	50–70 minutes
You need	Arduino Mega 2560, known-good USB data cable, Arch Linux host, `arduino-cli`, and this ADK checkout
Proof	A recorded output trace, three visible cycles, and a measured or inspected high-impedance shutdown
New words	digital level, output driver, high impedance, ownership, RAII

Safety boundary

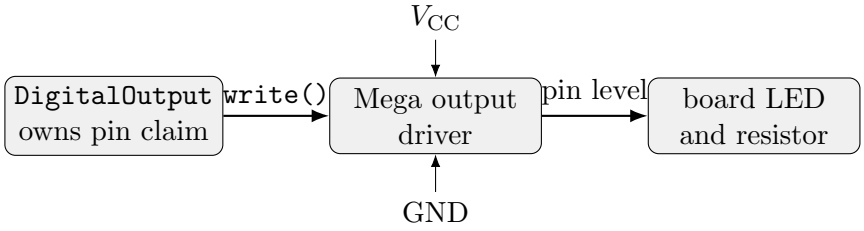
Use only the board's factory-installed LED and USB power. Work dry, on a nonconductive surface. Disconnect USB, barrel power, `VIN`, batteries, shields, and every lead before inspection. Proceed only when the board and cable are undamaged and no header pins touch.

Stop immediately for heat, smoke, odor, unexpected power flicker, repeated USB disconnects, or a loose connector. Disconnect upstream power. Do not touch a hot board or retry a damaged one. Label it DO NOT USE.

Check — all power removed before inspection
☐ dry board ☐ clear headers ☐ sound cable ☐ no other power source

1. Separate a command from an electrical state

A digital output actively drives a pin near ground for `Low` or near the board supply for `High`. High impedance, written `HiZ`, means the output driver is disconnected. It is not another voltage level. An unconnected high-impedance pin may float.

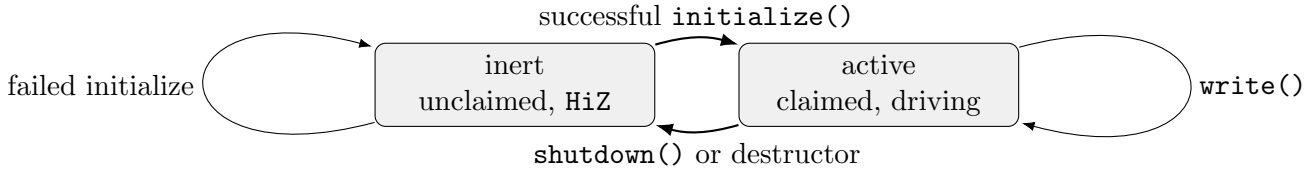


Exact lesson path: software object → `LED_BUILTIN` output driver → factory LED circuit. No external wire is added.

Predict. Which state actively sources or sinks current? _____ Which state releases the pin? _____

2. Read the ownership contract

ADK treats the pin as a resource, not a global integer. Construction produces an inert object. `initialize()` claims the pin and configures it transactionally. `write()` changes only an initialized, owned output. `shutdown()` is idempotent, selects input mode, and releases the pin as high impedance. Destruction calls shutdown, so stack unwinding in an exception-enabled host still cleans up even though ADK does not throw internally.



Operation	Success	Failure or repetition
construct	inert; no hardware access	not applicable
`initialize()`	claim, initial level, output mode	invalid or busy; remain inert
`write(level)`	drive requested level	report status; preserve contract
`shutdown()`	input mode, `HiZ`, release	repeated call is harmless
destruct	performs shutdown	never throws

3. Inspect the experimental first-class interface

This interface is host-verified but still experimental; it is not yet a stable supported release. `Runtime` owns the resource registry shared by components. The canonical complete example is `Lesson001DigitalOutput.ino` in its matching `examples` directory.

```
adk::Runtime       runtime;
adk::DigitalOutput output (runtime.resources (), LED_BUILTIN,
                           adk::Level::Low);

adk::Status status = output.initialize ();
if (!status.ok ())
{
    // Stay inert. Report the explicit status.
}

status = output.write (adk::Level::High);
status = output.write (adk::Level::Low);
output.shutdown ();
```

Why no implicit setup? A constructor cannot return a status without exceptions. Explicit initialization makes an invalid pin or resource conflict observable. RAII still guarantees cleanup after successful acquisition.

Why an explicit initial level? Setting the output latch before enabling the output driver avoids a brief unintended pulse. For the board LED, inactive is low. A generic output's safe electrical endpoint is high impedance.

4. Build evidence on the host first

The repository's fake Arduino functions record hardware calls as an ordered trace. Host tests are fast, deterministic, and require no connected board. This excerpt follows the assertions in `tests/test_io.cpp`.

```
fake::reset ();
adk::ResourceRegistry resources;
{
    adk::DigitalOutput output (resources, 13);

    require (output.initialize ().ok (),
             "output initialization");
    require (output.write (adk::Level::High).ok (),
             "output high write");
}
```

The test then checks this exact operation order: write `LOW`, select `OUTPUT`, write `HIGH`, and finally select `INPUT` during destruction. The claim is then available to a replacement object.

Check — test the boundaries, not only the happy path
☐ initialize success ☐ duplicate pin claim ☐ invalid pin ☐ write before initialize ☐ repeated shutdown ☐ destructor while active

Trace exercise. Write the expected operations when a second output tries to claim an already-owned pin:

5. Compile for the Mega 2560

From the repository root, run the documented checks and the lesson's Arduino compile target. Read the command before executing it. Confirm the fully qualified board name is `arduino:avr:mega`; never infer a

connected board or port.

```
make check
make arduino
arduino-cli board list
```

Record the compiler result and reported flash/RAM use:

Host tests

Mega compile

Flash bytes

Static RAM bytes

6. Run the board acceptance procedure

1. With all power removed, repeat the visual inspection.
2. Connect only the USB data cable.
3. Run `arduino-cli board list`. Copy the Mega's exact port.
4. Upload the lesson sketch using the repository's documented target. Do not paste a port belonging to another device.
5. Observe at least three complete cycles without touching conductors.
6. Invoke or wait for the sketch's documented shutdown phase.
7. Verify the LED becomes inactive. If instrumentation is available, measure pin mode safely; otherwise inspect the recorded host trace.
8. Disconnect USB at the computer end.

Check — hardware record
Port: _____ Board ID: _____
Three cycles: ☐ inactive at shutdown: ☐ unexpected behavior: ☐

Observation, not interpretation:

_____

_____

Claim supported by the observation and trace:

_____

_____

7. Debug with the smallest useful signal

Digital output comes first because one spare pin can expose control flow without a debugger. Assign meanings before running:

Pattern	Meaning
steady inactive	waiting or safely stopped
one short pulse	update loop reached
two short pulses	input event accepted
slow repeated pulse	recoverable status
fast repeated pulse	stop and inspect

Patterns are protocol, not decoration. Keep them deterministic, document the time base, and test the trace. Never use delay-based debug output in code whose timing contract forbids blocking.

8. Deliberately test misuse

Run these on the host fake. Do not create shorts or faults on hardware.

1. Construct two outputs for one pin. The second initialization must report a conflict without disturbing the first.
2. Call `write()` before initialization. Record its status and confirm no platform operation occurred.
3. Initialize with `NUM_DIGITAL_PINS`. Confirm `error() == StatusCode::InvalidPin`, an empty trace, and no resource claim.
4. Call `shutdown()` twice. Confirm the second call is harmless.
5. Leave scope while active. Confirm destructor cleanup matches explicit shutdown.

Explain. Why is "the LED went dark" weaker evidence than "the pin became high impedance"?

9. Extension: polarity belongs above the endpoint

`DigitalOutput` owns an electrical endpoint and speaks `High/Low`. A later `MonoLed` composes it and speaks `on/off`, accounting for active-high or active-low wiring. Do not teach a generic pin that `High` always means "on."

Exit ticket

1. What does the object own after successful initialization?
2. Why is `HiZ` not equivalent to `Low`?
3. Which operation must be safe to repeat?

4. How does explicit status reporting coexist with RAII?

5. Name one host-only fault test and its required final state.

Ready for lesson 002 when you can: ☐ predict the trace ☐ explain safe shutdown ☐ identify pin conflict ☐ compile for Mega ☐ separate electrical and semantic state

Further reading. Use the companion HTML API page for exact signatures, source links, current limitations, and searchable test names. Use this PDF as the bench worksheet. Arduino's Mega 2560 documentation supplies authoritative board limits; the ADK safety page supplies the project-wide stop conditions. Links are maintained in the HTML lesson to avoid stale printed URLs.